Causal Information Bottlenecks and Strategic Roadmap for Sub-15% Solar Irradiance Forecasting in POLARIS Stage 7

1. Diagnostic Forensic Assessment of Stage 6.6 Baseline and Unmodeled Physical Mechanisms

The operational benchmark for shortwave solar irradiance forecasting at the Nuolja research station (68.355701^\circ\text{N}, 18.763666^\circ\text{E}, 702\text{ m a.s.l.}) is established by the Stage 6.6 Liquid Water Path and Ice Water Path (LWP/IWP) Cloud Microphysics Champion. Evaluated on the held-out 173 ODD test origins under the frozen Stage 5.2 pipeline, the Stage 6.6 champion achieved a daytime Normalized Mean Absolute Error (\text{nMAE}) of 28.7026\%, corresponding to a daytime Mean Absolute Error (\text{MAE}) of 37.005888\text{ W/m}^2, a Root Mean Square Error (\text{RMSE}) of 65.671702\text{ W/m}^2, and a daytime prediction bias of +1.144965\text{ W/m}^2. This performance represents an absolute reduction of -2.078933\text{ W/m}^2 Daytime MAE (a relative improvement of -5.32\%) and -1.6125\text{ percentage points} nMAE relative to the Stage 5.2 baseline. A 10,000-sample origin-level paired bootstrap confirmed that this improvement is statistically significant, yielding a 95\% confidence interval of [-2.809838, -1.344760]\text{ W/m}^2 with zero bootstrap resamples crossing the null threshold.

Model Evaluation Stage	Daytime MAE (\text{W/m}^2)	Daytime nMAE (\%)	Daytime RMSE (\text{W/m}^2)	Daytime Bias (\text{W/m}^2)	Bootstrap 95% CI (\text{W/m}^2)	Operational Status
Stage 5.2 Control Baseline	39.084821	30.3151	68.993487	+1.934725	Reference Baseline	Retired Baseline
Stage 6.6 LWP/IWP Champion	**37.005888**	**28.7026**	**65.671702**	**+1.144965**	**[-2.8098, -1.3448]**	**Frozen Champion**
Absolute Improvement (\Delta)	**-2.078933**	**-1.6125\text{ pp}**	**-3.321785**	**-0.789760**	**100\% Negative**	**Statistically Significant**

The predictive gain realized in Stage 6.6 was achieved by addressing a physical misrepresentation of atmospheric cloud mass. By integrating 3\text{D} specific cloud liquid water (mass_fraction_of_cloud_condensed_water_in_air) and specific cloud ice (mass_fraction_of_cloud_ice_in_air) across 13 pressure levels (50\text{ hPa} to 1000\text{ hPa}) and truncating the vertical integration strictly at the model surface pressure (p_{\text{sfc}} \approx 891.8 \dots 974.1\text{ hPa}), Stage 6.6 provided the model with a direct proxy for total atmospheric optical column depth. Surface-aware boundary truncation eliminated non-physical

subterranean mass artifacts from the 1000\text{ hPa} level, allowing decision trees to distinguish between optically thin ice clouds (cirrus) and optically thick liquid clouds (stratus).
Conversely, the failure of earlier experimental tracks delineates clear boundaries regarding how atmospheric information must be structured:

1. **Spatial NWP Cloud Probes (Stage 5.4)**: Off-grid horizontal sampling at a 2.5\text{ km} spatial offset introduced variance noise without resolving local vertical optical columns, resulting in held-out test set degradation (\Delta\text{MAE} = +0.5804\text{ W/m}^2).
2. **MEPS Ensemble Spread (Stage 6.1)**: Scalar ensemble variance correlated strongly with error magnitude (\rho = +0.767) but overfit decision trees during cross-validation due to high-dimensional feature noise (\Delta\text{MAE} = +0.0457\text{ W/m}^2).
3. **Thermal-IR Satellite Brightness Temperature (Stage 6.2)**: Point/ROI thermal infrared observations (10.8\ \mu\text{m}) failed (\Delta\text{MAE} = +0.0290\text{ W/m}^2) because thermal skin temperature saturates at cloud top and cannot resolve vertical cloud thickness, cloud phase, or multi-layer cloud structures under high-latitude temperature inversions.
4. **Deterministic NWP Revision Trajectories (Stage 6.4)**: Update trajectories suffered from non-stationary issuance delays and failed fold stability criteria (+0.2857 W/m² degradation in Fold 1).

The remaining daytime error (\text{nMAE} = 28.7026\%) is governed by unmodeled atmospheric radiative transfer physics. At Arctic latitudes (68.35^\circ\text{N}), surface shortwave irradiance (SW_IN) is determined by the Beer-Lambert-Bouguer extinction law:
SW_IN = I_0 \cdot \cos(\theta_z) \cdot \exp\left(- \frac{\tau_{\text{Rayleigh}} + \tau_{\text{gases}} + \tau_{\text{aerosol}} + \tau_{\text{cloud}}}{\cos(\theta_z)} \right) + I_{\text{diffuse}}
The current model predicts total surface flux directly as an unconstrained scalar using gradient boosting trees (HistGradientBoostingRegressor). Five unmodeled physical mechanisms drive the remaining forecast error:

- **Uncoupled Direct and Diffuse Radiation Fluxes**: Total shortwave flux is treated as a monolithic variable. Cloud fields attenuate Direct Normal Irradiance (DNI) through single-scattering extinction while enhancing Diffuse Horizontal Irradiance (DHI) through forward scattering. Thin clouds convert direct beam flux into diffuse flux with minimal attenuation of global horizontal irradiance, whereas thick stratus reflects shortwave flux back to space.
- **Absence of Dynamic Aerosol Optical Depth (\tau_{\text{aerosol}})**: The current pipeline assumes a static clear-sky turbidity profile. Variable Arctic aerosol burdens, including sea salt, polar haze, and summer biomass burning plumes, alter clear-sky direct extinction and diffuse scattering.
- **Slant-Path Geometry at High Zenith Angles**: At Nuolja, high solar zenith angles (\theta_z > 70^\circ) extend atmospheric optical path lengths (m(\theta_z) \approx 1/\cos\theta_z) beyond 3 to 5 air masses. Consequently, spatial cloud structures located tens of kilometers up-sun dictate direct beam attenuation, whereas overhead vertical cloud columns control diffuse sky radiance.
- **Droplet Effective Radius (r_e) and Optical Depth non-Linearity**: Cloud optical depth (\tau_{\text{cloud}}) scales with liquid water path (LWP) and droplet effective radius (r_e) as \tau_{\text{cloud}} \approx \frac{3}{2} \frac{LWP}{\rho_w \cdot r_e}. Treating LWP as a linear feature ignores shifts in cloud particle size distributions.
- **Snow-Cloud Albedo Multiple Reflection Coupling**: High surface albedo (\alpha_{\text{surface}} > 0.8) from fresh snow surrounding Nuolja induces multiple

reflections between the ground and cloud bases, elevating surface diffuse irradiance relative to dark-surface assumptions.

2. Systematic Feasibility Investigation of Candidate Data Sources

To evaluate retrievable 2024 causal observations and NWP fields, candidate data sources were investigated across 14 data-engineering criteria.

Data Source 1: MEPS 3D Pressure-Level Full Hydrometeor and Cloud Profile Archive

The MEPS 3\text{D} pressure-level dataset provides vertical profiles of atmospheric state variables across 13 pressure levels (50\text{ hPa} to 1000\text{ hPa}) over a 2.5\text{ km} grid. Access is available through the MET Norway THREDDS server via OPeNDAP.

Data Source Attribute	Technical Characterization and Operational Specification
1. Dataset / Product Name	MEPS (MetCoOp Ensemble Prediction System) Deterministic 3D Pressure-Level Archive.
2. Exact Variables	specific_cloud_liquid_water_content, specific_cloud_ice_water_content, cloud_area_fraction_in_atmosphere_layer, relative_humidity, air_temperature, divergence_of_wind, air_pressure_at_cloud_base, air_pressure_at_cloud_top.
3. URL / Archive Location	https://thredds.met.no/thredds/catalog/meps25eps/catalog.html.
4. 2024 Availability	Complete 365-day historical archive for 2024.
5. Spatial Resolution	2.5\te[span_23](start_span)[span_23](end_span)xt{ km} horizontal grid (Equidistant Conic Projection).
6. Temporal Resolution	Hourly output for 00, 06, 12, 18 UTC initialization runs.
7. Vertical Resolution	13 pressure levels: 50, 100, 150, 200, 250, 300, 400, 500, 700, 800, 850, 925, 1000 hPa.
8. Historical Accessibility	Accessible via NetCDF subsetting and OPeNDAP protocol on MET Norway THREDDS.
9. Causal Availability at t_0	Available at t_0 for runs initialized at or prior to t_0 (t_{\text{ref}} \le t_0 - 2.5\text{h} operational delivery).
10. Extraction Method	OPeNDAP point-profile slice centered at Nuolja grid coordinates (y=760, x=486).
11. Storage / Compute Cost	Low storage footprint (< 1.5\text{ GB} for full 2024 point-extracted profile history); minimal

Data Source Attribute	Technical Characterization and Operational Specification
	CPU overhead.
12. POLARIS Tested Status	Partially tested in Stage 6.6 (vertically integrated LWP/IWP). **Untested**: Layer-by-layer cloud fraction, vertical humidity gradients, discrete cloud base/top pressures.
13. Expected Physical Value	High. Provides vertical cloud layer distributions, enabling the identification of overlapping cloud decks and boundary layer cloud base heights.
14. Leakage Risks	Zero risk if t_{\text{ref}} < t_0 is enforced.

Data Source 2: MEPS Surface Direct and Diffuse Radiation Diagnostics

MEPS outputs accumulated surface shortwave and longwave radiation fluxes. Extracting separate direct and diffuse downwelling shortwave components isolates the NWP radiative transfer model's explicit partitioning of beam extinction versus atmospheric scattering.

Data Source Attribute	Technical Characterization and Operational Specification
1. Dataset / Product Name	MEPS Operational Surface Radiation Diagnostics.
2. Exact Variables	surface_direct_downwelling_shortwave_flux_in_air, integral_of_surface_downwelling_shortwave_flux_in_air_wrt_time, integral_of_surface_downwelli[span_4](start_span)[span_4](end_span)ng_longwave_flux_in_air_wrt_time.
3. URL / Archive Location	MET Norway OpenData / THREDDS MEPS Post-Processed Catalogue.
4. 2024 Availability	Complete 2024 coverage.
5. Spatial Resolution	2.5\text{ km} horizontal grid.
6. Temporal Resolution	Hourly accumulated fluxes (\text{W s m}^{-2}) converted to average hourly fluxes (\text{W/m}^2).
7. Vertical Resolution	Surface diagnostic variables (0\text{ m AGL}).
8. Historical Accessibility	Direct OPeNDAP tabular extraction via THREDDS.
9. Causal Availability at t_0[span_6](start_span)[span_6](end_span)	Available at t_0 for NWP forecast lead times H1 \dots H24.
10. Extraction Method	Numerical differentiation of hourly accumulated integrals at grid point (y=760, x=486).
11. Storage / Compute Cost	Minimal storage (< 100\text{ MB}); lightweight computation.
12. POLARIS Tested Status	**Never Tested**. POLARIS currently uses Open-Meteo total cloud cover and GHI. Raw

Data Source Attribute	Technical Characterization and Operational Specification
	MEPS direct vs. diffuse flux components have not been extracted.
13. Expected Physical Value	High. Isolates the NWP radiative transfer scheme's explicit partitioning of direct beam extinction versus diffuse scattering.
14. Leakage Risks	Zero risk; valid future forecast covariates.

Data Source 3: CAMS Global Atmospheric Composition Forecasts

The Copernicus Atmosphere Monitoring Service (CAMS) provides global forecasts of Aerosol Optical Depth (AOD) at 550\text{ nm} along with individual aerosol species components. This dataset fills the clear-sky turbidity information gap in POLARIS.

Data Source Attribute	Technical Characterization and Operational Specification
1. Dataset / Product Name	Copernicus Atmosphere Monitoring Service (CAMS) Global Atmospheric Composition Forecasts.
2. Exact Variables	aod550 (Total Aerosol Optical Depth at 550nm), duaod550 (Dust AOD), ssaod550 (Sea Salt AOD), omaod550 (Organic Matter AOD), suaod550 (Sulphate AOD), bcaod550 (Black Carbon AOD), tcwv (Total Column Water Vapour).
3. URL / Archive Location	Copernicus Atmosphere Data Store (ADS): https://ads.atmosphere.copernicus.eu.
4. 2024 Availability	Complete 2024 historical forecast archive available.
5. Spatial Resolution	0.4^\circ \times 0.4^\circ (\approx 40\text{ km} horizontal resolution).
6. Temporal Resolution	3-hourly forecast steps initialized daily at 00:00 UTC and 12:00 UTC.
7. Vertical Resolution	Integrated total column attributes.
8. Historical Accessibility	Python cdsapi batch retrieval from ADS.
9. Causal Availability at t_0	Available at t_0 from the 00:00 UTC CAMS run delivered by 06:00 UTC.
10. Extraction Method	Bilinear interpolation to Nuolja station coordinates (68.3557^\circ\text{N}, 18.7637^\circ\text{E}).
11. Storage / Compute Cost	Negligible (< 50\text{ MB} for 2024 time series); vector tabular join.
12. POLARIS Tested Status	**Never Tested**. POLARIS currently assumes static clear-sky aerosol turbidity.
13. Expected Physical Value	High during clear-sky and thin-cloud regimes. Eliminates bias caused by variable aerosol scattering during Arctic summer months.

Data Source Attribute	Technical Characterization and Operational Specification
14. Leakage Risks	Zero risk if CAMS forecast issuance time precedes t_0.

Data Source 4: EUMETSAT NWC SAF / PPS Polar-Orbiting Satellite Cloud Product Retrievals

The Nowcasting SAF Polar Processed System (NWC SAF PPS) processes AVHRR, VIIRS, and MODIS radiances from polar-orbiting satellites (MetOp, NOAA, Suomi-NPP), retrieving physical cloud parameters over High-Latitude regions.

Data Source Attribute	Technical Characterization and Operational Specification
1. Dataset / Product Name	EUMETSAT NWC SAF PPS Polar-Orbiting Satellite Cloud Microphysics Products.
2. Exact Variables	cloud_optical_thickness (COT), cloud_top_height (CTH), cloud_top_temperature (CTT), cloud_phase (CP), cloud_liquid_water_path (CLWP), cloud_drop_effective_radius (r_e).
3. URL / Archive Location	MET Norway Satellite Data Archive / THREDDS: https://thredds.met.no/thredds/catalog/sat/catalog.html.
4. 2024 Availability	Complete coverage over northern Scandinavia via MetOp-B/C, NOAA-19/20, Suomi-NPP satellite overpasses.
5. Spatial Resolution	1\text{ km} \times 1\text{ km} at nadir.
6. Temporal Resolution	Swath-based orbital overpasses (14–18 overpasses daily at 68^\circ\text{N}; sampling interval 40–90 minutes).
7. Vertical Resolution	Column-integrated cloud properties anchored at cloud top height.
8. Historical Accessibility	NetCDF swath file retrieval via MET Norway THREDDS satellite catalogue.
9. Causal Availability at t_0	Strict observational cut-off: satellite overpass time t_{\text{overpass}} \le t_0 - 30\text{ minutes} processing latency.
10. Extraction Method	Nearest-neighbor spatial aggregation within a 5\text{ km} radius centered at Nuolja.
11. Storage / Compute Cost	Moderate (< 10\text{ GB} aggregated time-series data); requires orbit-to-grid mapping.
12. POLARIS Tested Status	**Never Tested**. Stage 6.2 tested basic thermal-IR channels. Advanced multi-spectral physical retrievals (COT, CWP, r_e) were not tested.

Data Source Attribute	Technical Characterization and Operational Specification
13. Expected Physical Value	High for initial forecast horizons (H1\dots H6). Provides true observational cloud optical thickness at forecast issuance time t_0.
14. Leakage Risks	High if swath overpasses post-dating t_0 are included. Requires strict timestamp filtering (t_{\text{observation}} \le t_0).

Data Source 5: MET Nordic High-Resolution Gridded Operational Analysis V4

MET Nordic V4 integrates MEPS NWP outputs with surface station observations and radar datasets to generate 1\text{ km} gridded analyses. This dataset provides terrain-adjusted surface meteorological context surrounding the station prior to forecast issuance.

Data Source Attribute	Technical Characterization and Operational Specification
1. Dataset / Product Name	MET Nordic Gridded Analysis / Reanalysis Version 4.
2. Exact Variables	air_temperature_2m, relative_humidity_2m, surface_air_pressure, precipitation_amount, integral_of_surface_downwelling_shortwave_flux_in_air_wrt_time.
3. URL / Archive Location	https://thredds.met.no/thredds/catalog/metnordic/catalog.html.
4. 2024 Availability	Complete 2024 historical archive.
5. Spatial Resolution	1\text{ km} \times 1\text{ km} high-resolution terrain-adjusted grid.
6. Temporal Resolution	Hourly observational analysis updates.
7. Vertical Resolution	Surface / 2\text{ m} height adjustments based on high-resolution DEM.
8. Historical Accessibility	HTTPS / OPeNDAP slice access from MET Norway archive.
9. Causal Availability at t_0	Available for historical context t \le t_0 - 1\text{ hour}.
10. Extraction Method	Spatial point extraction at Nuolja coordinates.
11. Storage / Compute Cost	Minimal for point extraction (< 200\text{ MB}).
12. POLARIS Tested Status	**Never Tested**. POLARIS currently relies solely on SITES station observations for ground context.
13. Expected Physical Value	Moderate. Provides spatial elevation-corrected meteorological surface context surrounding the station prior to forecast issuance.
14. Leakage Risks	Moderate risk if analysis timestamps post-dating t_0 are included. Requires strict filtering (t_{\text{analysis}} < t_0).

3. Comparative Evaluation of Model Formulations

To overcome the architectural limits of decision-tree regression (HistGradientBoostingRegressor) observed in Stages 5.2 through 6.6, nine candidate model formulations were evaluated.

```
Direct SW_IN Regression (A)  --> Predicts raw W/m² directly (Current Baseline)
Clear-Sky Index k_t (B)      --> Predicts k_t = SW_IN / GHI_clearsky in [0, 1.2]
Physics + ML Hybrid (C)      --> ML predicts optical depth τ; Differentiable Two-Stream RT Solver outputs flux
Horizon-Specific Models (D) --> 24 independent estimators for H1...H24
State-Space Sequences (E)    --> Temporal Fusion Transformers over sequence histories
Regime Transitioning (F)     --> Hidden Markov / Regime-switching Correctors (Rejected Stage 4.4)
Probabilistic Quantiles (G)  --> Conformalized Quantile Boosting (q0.05...q0.95)
Multi-Task Predictions (H)  --> Joint prediction of DNI, DHI, and k_t with shared latent representation
Physics-Guided Loss (I)      --> GBDT/NN trained with energy-conservation penalty loss
```

The tree-based decision models currently utilized split feature space into orthogonal hyper-rectangles. This architecture exhibits three key failure modes:

1. It cannot extrapolate beyond the maximum irradiance values observed in the training set.
2. It generates step-function outputs that create artificial discontinuities along smooth diurnal solar curves.
3. It conflates astronomical solar geometry with atmospheric cloud attenuation, forcing the model to re-learn seasonal solar elevation patterns from data.

Formulation Code & Name	Primary Mathematical Mechanism	Ability to Overcome Decision-Tree Limitations	Physical Exploitation Potential
A. Direct SW_IN Regression	Direct mapping: \hat{y} = f(X) \to \text{W/m}^2.	Low. Subject to step-function artifacts and seasonal scale drift.	Baseline Reference
B. Clear-Sky-Index (k_t) Prediction	Predicts k_t = \frac{SW_IN}{GHI_{\text{clearsky}}}; outputs \hat{y} = \hat{k}_t \cdot GHI_{\text{clearsky}}.	High. Factors out solar geometry; bounds target in [0, 1.2].	Very High
C. Physics + ML Hybrid (Delta-Eddington)	ML predicts layer optical depth \tau; analytical two-stream solver computes fluxes.	High. Guarantees physical energy conservation and smooth diurnal curves.	Extreme
D. Horizon-Specific	24 independent	Moderate. Eliminates	Moderate

Formulation Code & Name	Primary Mathematical Mechanism	Ability to Overcome Decision-Tree Limitations	Physical Exploitation Potential
Models (H1 \dots H24)	estimators fit to individual lead times h \in \{1 \dots 24\}.	horizon feature blending but increases parameter footprint 24-fold.	
E. Sequence / State-Space Models (TFT)	Attention mechanisms over historical context (24\text{h}) and future NWP sequences.	High. Learns temporal dynamics and cloud front propagation.	High
F. Cloud-State Transition Models	Regime classification followed by state-dependent corrections.	Low. Discrete state switches create boundary instability (Failed Stage 4.4).	Low (Rejected)
G. Probabilistic Quantile Forecasting	Pinball loss quantile estimation (q_{0.05} \dots q_{0.95}) with Conformal Calibration.	Moderate. Quantifies uncertainty for energy dispatch; does not improve point MAE.	High (for EMS)
H. Multi-Task Latent Prediction	Joint prediction of DNI, DHI, and k_t via shared neural representation.	High. Forces latent features to maintain physical consistency across radiation components.	High
I. Physics-Guided Loss Functions	Loss function augmented with Beer-Lambert penalty: L = L_{\text{MAE}} + \lambda \cdot L_{\text{physics}}.	Moderate. Penalizes physical violations during GBDT split selection.	Moderate

4. Strategic Candidate Synthesis and Priority Ranking

The 10 Strongest Candidate Directions

1. **Clear-Sky-Index (k_t) Transmittance Target Transformation**: Reformulating the prediction target as non-dimensional atmospheric transmittance k_t = SW_IN / GHI_{\text{clearsky}} isolates atmospheric attenuation from astronomical solar geometry.
2. **Explicit MEPS Direct and Diffuse Radiation Decomposition**: Extracting raw NWP shortwave components (surface_direct_downwelling_shortwave_flux_in_air) separates direct beam extinction from diffuse sky scattering.
3. **CAMS Dynamic Aerosol Optical Depth (AOD_{550}) Integration**: Incorporating 3-hourly dynamic aerosol optical depth fields eliminates clear-sky turbidity errors during Arctic summer months.
4. **Satellite Polar Cloud Microphysics Retrievals (NWC SAF / PPS)**: Integrating Cloud Optical Thickness (COT) and Cloud Phase (CP) from polar satellite overpasses provides observational cloud state at forecast issuance time t_0.

5. **Differentiable Two-Stream Radiative Transfer Hybrid Modeling**: Coupling machine learning predictions of optical depth (\tau[span_12](start_span)[span_12](end_span)) with an analytical Delta-Eddington radiative transfer solver guarantees physical energy conservation.
6. **Multi-Layer Vertical Cloud Fraction Profiles**: Extracting discrete layer cloud fractions across high, mid, and low pressure levels resolves overlapping cloud decks.
7. **Slant-Path Optical Path Length Adjustment**: Recomputing cloud liquid water path along the solar elevation vector (LWP_{\text{slant}} = LWP / \cos\theta_z) accounts for high solar zenith angles.
8. **Multi-Refine Snow-Cloud Albedo Coupling**: Incorporating surface snow depth (D_SNOW) and satellite surface albedo models multiple reflections between ground and cloud bases.
9. **Multi-Task Latent Representation Learning**: Jointly predicting DNI, DHI, and k_t using a shared neural architecture enforces physical consistency across shortwave components.
10. **Conformalized Probabilistic Quantile Estimation**: Outputting calibrated quantile bounds (q_{0.05} \dots q_{0.95}) supplies downstream energy management engines with uncertainty bounds.

The 5 Most Promising Actual Data Sources

1. **MEPS Surface Direct and Diffuse Radiation Diagnostics**: Direct extraction of raw NWP direct and diffuse shortwave components.
2. **CAMS Global Atmospheric Composition Forecasts**: CAMS 3-hourly AOD_{550} and aerosol species composition.
3. **MEPS 3D Pressure-Level Hydrometeor and Cloud Fields**: Full 13-layer pressure-level cloud fraction and humidity profiles.
4. **EUMETSAT NWC SAF PPS Polar-Orbiting Satellite Retrievals**: Physical Cloud Optical Thickness (COT) and Cloud Water Path (CWP) retrievals.
5. **MET Nordic Gridded Analysis V4**: 1\text{ km} terrain-adjusted surface meteorological analyses.

The 3 Most Promising Modeling Formulations

1. **Clear-Sky-Index (k_t) Transmittance Target Formulation**: Normalizes solar geometry out of the prediction target, converting non-stationary irradiance into a bounded transmittance index k_t \in [0, 1.2].
2. **Physics + ML Hybrid (Delta-Eddington Radiative Transfer)**: Couples ML-predicted optical depth (\tau_{\text{cloud}}) with an analytical two-stream radiative transfer solver, guaranteeing physical consistency and energy conservation.
3. **Multi-Task Deep Learning Representation**: Jointly predicts DNI, DHI, and k_t using a shared latent representation, preventing physical contradictions between predicted radiation components.

The Single Most Promising Overall Route

The single most promising route to approach the target is the **Clear-Sky-Index Transmittance Modeling (k_t) Architecture coupled with MEPS Direct/Diffuse Flux Decomposition and CAMS Dynamic Aerosol Optical Depth (AOD_{550})**.

This route directly addresses the core physical failure modes of Stage 6.6 without introducing spatial mismatch or violating causal locks:

1. Normalizing the target variable by GHI_{\text{clearsky}} eliminates astronomical solar geometry error from the loss function, converting the target into a stationary transmittance index bounded in [0, 1.2].
2. Incorporating MEPS direct and diffuse flux diagnostics provides explicit NWP radiative transfer information.
3. Incorporating CAMS AOD_{550} eliminates clear-sky turbidity bias during summer months.

5. Stage 7 Experimental Sequence and Execution Protocol

The proposed experimental sequence strictly preserves the locked benchmark protocol: 347 clean origins, 174 EVEN origins reserved for cross-validation, and 173 ODD origins locked as an untouched test set.

```
Stage 6.6 Frozen Champion Baseline (EVEN CV MAE: 38.2806 W/m², ODD
nMAE: 28.7026%)
                              |
                              v
Experiment 7.1: MEPS Direct and Diffuse Shortwave Flux Decomposition
   - Extract raw MEPS direct/diffuse fluxes; evaluate on EVEN 5-Fold
C[span_14](start_span)[span_14](end_span)V
   - Acceptance: EVEN CV MAE <= 37.7806 W/m² (Delta >= -0.50 W/m²),
4/5 folds improved
                              |
                              v
Experiment 7.2: CAMS Dynamic Aerosol Optical Depth (AOD_550)
Integration
   - Add CAMS
AO[span_86](start_span)[span_86](end_span)[span_91](start_span)[span_9
1](end_span)D550 and aerosol species attenuation
factors[span_94](start_span)[span_94](end_span)[span_95](start_span)[s
pan_95](end_span)
   - Acceptance: EVEN CV MAE <= 37.4806 W/m² (Cumulative
Delt[span_44](start_span)[span_44](end_span)a >= -0.80 W/m²)
                              |
                              v
Experiment 7.3: Clear-Sky-Index (k_t) Transmittance Target Formulation
   - Transform target to k_t = SW_IN / GHI_clearsky; train GBDT on k_t
   - Acceptance: EVEN CV MAE <= 36.5000 W/m²
([span_15](start_span)[span_15](end_span)Cumulative Delta >= -1.78
W/m², nMAE < 27.5%)
                              |
                              v
Experiment 7.4: Differentiable Delta-Eddington Two-Stream Physics
Hybrid
```

```
   - ML predicts optical depth profiles; analytical two-stream solver
outputs fluxes
   - Acceptance: EVEN CV MAE <= 34.5000 W/m² (nMAE < 25.0%) -> Trigger
ODD Evaluation Gate
```

Experiment 7.1: MEPS Direct and Diffuse Shortwave Flux Decomposition

- **New Information**: Explicit partitioning of surface shortwave radiation into direct beam flux (surface_direct_downwelling_shortwave_flux_in_air) and diffuse flux.
- **Data Source**: MEPS Post-Processed Surface Radiation Diagnostics Archive on MET Norway THREDDS.
- **Causal Rule**: Extracted for forecast horizons H1 \dots H24 using NWP runs initialized prior to t_0 (t_{\text{ref}} \le t_0 - 2.5\text{h}).
- **Feature Representation**: meps_sw_direct, meps_sw_diffuse, meps_direct_ratio = meps_sw_direct / (meps_sw_direct + meps_sw_diffuse + 1e-5).
- **Model / Formulation**: HistGradientBoostingRegressor (matching Stage 6.6 champion parameters: depth 4, min_samples_leaf 30, l2 10.0) predicting SW_IN residual.
- **EVEN-CV Test**: 5-fold forward-chaining cross-validation across 174 EVEN origins.
- **Acceptance Criteria**: Aggregate EVEN-CV Daytime MAE \le 37.7806\text{ W/m}^2 (\Delta\text[span_16](start_span)[span_16](end_span){MAE} \le -0.50\text{ W/m}^[span_42](start_span)[span_42](end_span)2 vs Stage 6.6 CV baseline of 38.2806\text{ W/m}^2); improvement in \ge 4/5 folds; max single-fold degradation \le +0.15\text{ W/m}^2.
- **Likely Failure Mode**: NWP direct/diffuse partitioning at 2.5\text{ km} resolution may exhibit systematic spatial biases over mountainous terrain.

Experiment 7.2: CAMS Dynamic Aerosol Optical Depth (AOD_{550}) Extinction Integration

- **New Information**: Dynamic 3-hourly atmospheric aerosol optical depth and species breakdown.
- **Data Source**: Copernicus Atmosphere Data Store (ADS) CAMS Global Atmospheric Composition Forecasts.
- **Causal Rule**: CAMS forecast initialized at 00:00 UTC, delivered by 06:00 UTC, matched causally to origins t_0 \ge 06:00\text{ UTC}.
- **Feature Representation**: cams_aod550, cams_dust_aod, cams_sulphate_aod, aerosol_attenuation_factor = exp(-cams_aod550 / cos(solar_zenith)).
- **Model / Formulation**: HistGradientBoostingRegressor integrated with Stage 6.6 features.
- **EVEN-CV Test**: 5-fold forward-chaining CV on 174 EVEN origins.
- **Acceptance Criteria**: Aggregate EVEN-CV Daytime MAE \le 37.4806\text{ W/m}^2 (\Delta\text{MAE} \le -0.80\text{ W/m}^2 cumulative); improvement in \ge 4/5 folds.
- **Likely Failure Mode**: 0.4^\circ horizontal resolution of CAMS may be too coarse to capture local aerosol variations at Nuolja.

Experiment 7.3: Clear-Sky-Index (k_t) Transmittance Target

Formulation

- **New Information**: Reformulation of the prediction target to atmospheric transmittance k_t = SW_IN / GHI_{\text{clearsky}}, decoupling astronomical geometry from cloud attenuation.
- **Data Source**: SITES ground truth observations combined with PVLIB clear-sky GHI modeling.
- **Causal Rule**: Target derived strictly from ground truth SW_IN(t_0+h) and deterministic solar geometry at valid time t_0+h.
- **Feature Representation**: All Stage 6.6 features plus features from Exp 7.1 and 7.2. Target: k_t = SW_IN / \max(GHI_{\text{clearsky}}, 5.0). Post-processing: \widehat{SW_IN} = \hat{k}_t \cdot GHI_{\text{clearsky}}, masked to 0.0\text{ W/m}^2 when \theta_{\text{elevation}} \le 0^\circ.
- **Model / Formulation**: HistGradientBoostingRegressor trained on target k_t using absolute error loss.
- **EVEN-CV Test**: 5-fold forward-chaining CV on 174 EVEN origins.
- **Acceptance Criteria**: Aggregate EVEN-CV Daytime MAE \le 36.5000\text{ W/m}^2 (\Delta\text{MAE} \le -1.78\text{ W/m}^2 cumulative); relative EVEN nMAE < 39.0\%.
- **Likely Failure Mode**: Low clear-sky GHI values at low solar elevation angles (\theta_{\text{elevation}} < 5^\circ) may cause instability in k_t, requiring smooth thresholding near zero.

Experiment 7.4: Differentiable Two-Stream Radiative Transfer Hybrid Model

- **New Information**: Analytical two-stream shortwave radiation solver embedded directly into the machine learning prediction pipeline.
- **Data Source**: MEPS 3D cloud liquid/ice profiles combined with SITES ground observations.
- **Causal Rule**: Strict adherence to t_{\text{ref}} < t_0 for all input fields.
- **Feature Representation**: ML model predicts layer cloud optical depth (\tau_i) for 13 pressure levels; two-stream solver computes surface direct and diffuse transmittance.
- **Model / Formulation**: PyTorch-based neural hybrid trained end-to-end with L_1 loss on surface SW_IN.
- **EVEN-CV Test**: 5-fold forward-chaining CV on 174 EVEN origins.
- **Acceptance Criteria**: Aggregate EVEN-CV Daytime MAE \le 34.5000\text{ W/m}^2 (\Delta\text{MAE} \le -3.78\text{ W/m}^2 vs Stage 6.6); ODD set remains strictly locked.
- **Likely Failure Mode**: Gradient instability during backpropagation through the two-stream matrix solver under extreme optical depth conditions (\tau > 100).

6. Synthesis and Strategic Outlook

Achieving a daytime nMAE below 15\% requires moving beyond direct scalar irradiance regression on unconstrained tree ensembles. The Stage 6.6 LWP/IWP champion confirmed that incorporating vertical atmospheric structure provides meaningful predictive signal. The Stage 7 roadmap addresses remaining radiative transfer errors by reformulating the prediction target to atmospheric transmittance (k_t), incorporating explicit MEPS direct and diffuse shortwave flux

components, and integrating dynamic CAMS aerosol optical depth. This sequence establishes a physically grounded path toward sub-15% solar irradiance forecasting while preserving benchmark integrity and causal compliance.

Works cited

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